

CONSTRUCTION MANAGEMENT SYSTEMS — SCHEDULE CONTROL DEEP DIVE

CPM SCHEDULING

THE FINANCIAL WEAPON

MOST GCS NEVER LEARN

SUBTITLE_SPEC_01

Why Your Schedule Is Either Making You Money or Losing It

MOST GCS DON'T HAVE A SCHEDULE

THEY HAVE A WISH LIST

STATUS: INADEQUATE

WHAT MOST GCS CALL A SCHEDULE

- ✗ ~~A bar chart with dates~~
- ✗ ~~No logic ties between activities~~
- ✗ ~~Built once, never updated~~
- ✗ ~~Useless on the jobsite~~

STATUS: REQUIRED

WHAT A SCHEDULE ACTUALLY IS

- ✓ A logic-driven financial model
- ✓ Every activity linked by dependency
- ✓ Updated weekly to reflect reality
- ✓ The backbone of project control

THE
COST



"Without real CPM: reactive decisions, blown margins, zero leverage in disputes.
You're not managing – you're guessing."

WHAT IS CPM?

CRITICAL PATH METHOD — IN PLAIN ENGLISH

DEFINITION



The **longest chain** of dependent activities that determines your project end date. Not the most work — the longest linked sequence.

THE RULE



If any activity on the critical path slips, the **entire project slips**. No exceptions.

FLOAT



Non-critical activities have float — they can slip without moving the end date. **Critical path activities have zero float.**

WHY IT MATTERS



CPM tells you **WHERE to focus** and what you can safely ignore. It replaces gut with math.

HARD TRUTH

"IF YOU DON'T KNOW WHICH ACTIVITIES ARE ON YOUR CRITICAL PATH, EVERY ACTIVITY FEELS URGENT. **AND WHEN EVERYTHING IS URGENT, NOTHING GETS MANAGED.**"

THE 5 BUILDING BLOCKS

EVERY CPM SCHEDULE IS BUILT FROM THESE COMPONENTS



01

ACTIVITIES

What gets done. Every task on the project, clearly defined.



02

DURATIONS

How long each activity takes. Based on crew size, scope, and conditions.



03

LOGIC / DEPENDENCIES

What has to happen before what. The relationships between activities.



04

RESOURCES

Who does the work. Crews, subs, equipment, inspectors.



05

CONSTRAINTS

Hard dates, weather windows, inspections, client decisions. Things you can't move.

// REMOVE ANY ONE OF THESE AND YOU DON'T HAVE A SCHEDULE. YOU HAVE A LIST.

ACTIVITY BREAKDOWN

IF YOU CAN'T DEFINE IT, YOU CAN'T SCHEDULE IT

// WORK BREAKDOWN STRUCTURE (WBS)

└ 1. PROJECT SCOPE

└ 1.1 PHASES

└ 1.1.1 ACTIVITIES

EACH ACTIVITY MUST BE:

- ✓ Measurable
- ✓ Assignable
- ✓ Clear Start/Finish Criteria

// EXAMPLE: OUTDOOR LIVING PROJECT

BAD PRACTICE

~~"Do electrical"~~

Too broad. Unmeasurable. Unassignable.



GOOD PRACTICE

- > Electrical rough-in to pergola (2 days)
- > Low-voltage lighting conduit (1 day)
- > Fixture installation (1 day)
- > Final electrical inspection (0.5 day)



"The #1 scheduling mistake: activities too broad to manage. If you can't tell when it started and when it finished, it's not an activity – it's a wish."

DEPENDENCIES

LOGIC IS EVERYTHING

FS FINISH-TO-START

MOST COMMON



Activity B can't start until Activity A finishes.

| Ex: "Can't pour concrete until forms are set."

FF FINISH-TO-FINISH



Activity B can't finish until Activity A finishes.

| Ex: "Punch list can't close until final inspection closes."

SS START-TO-START



Activity B can start when Activity A starts.

| Ex: "Framing and rough electrical can start together."

SF START-TO-FINISH

RARE



Activity B can't finish until Activity A starts.

| Almost never used in construction.



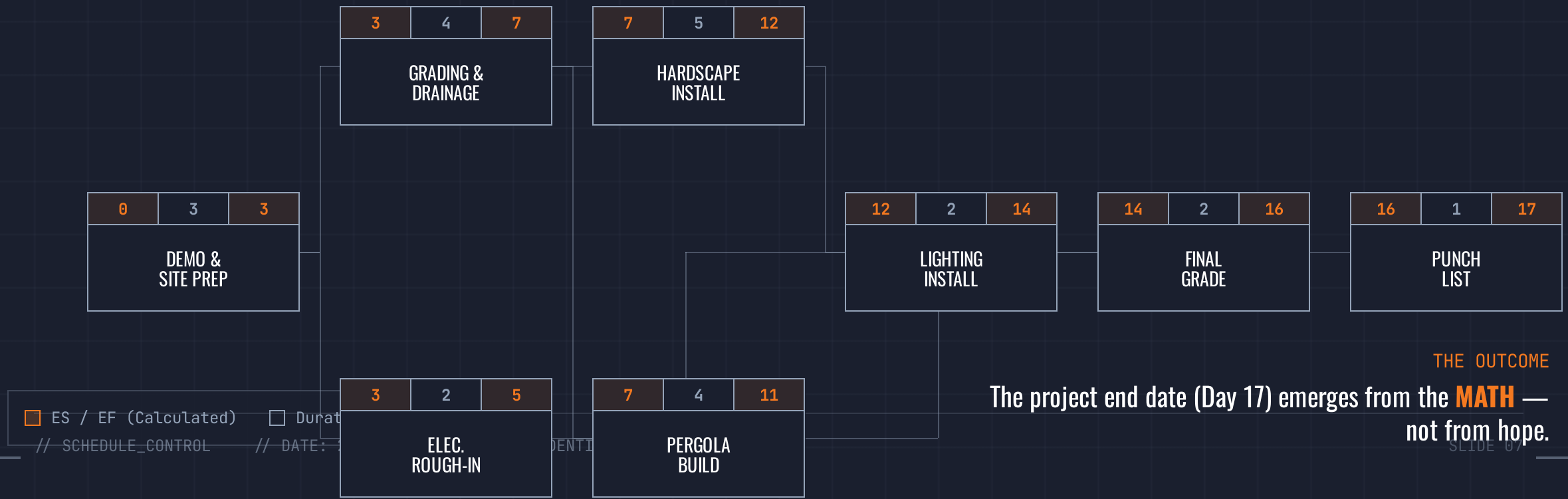
THE PM MANDATE

"ASK: 'WHAT MUST BE DONE FIRST?' BAD LOGIC = FAKE SCHEDULE."

THE FORWARD PASS

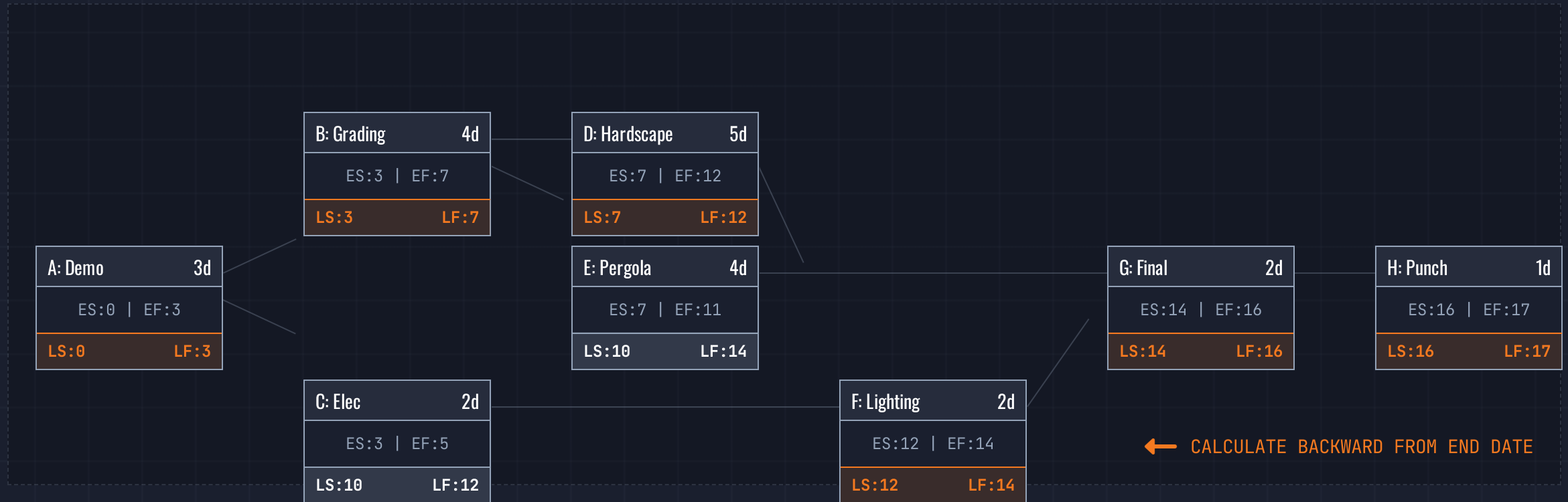
EARLIEST EVERYTHING CAN HAPPEN

CALCULATION FLOW →



THE BACKWARD PASS

LATEST EVERYTHING CAN HAPPEN **WITHOUT SLIPPING**



THE FORMULA

$$LS = LF - \text{Duration}$$

Start from the end, subtract duration.

KEY INSIGHT

"The backward pass reveals which activities have **NO room to slip**. When $LS = ES$, that activity is critical."

FLOAT (TOTAL FLOAT)

YOUR MARGIN OF ERROR ON EVERY ACTIVITY

THE MATH

$$\text{FLOAT} = \text{Late Start} - \text{Early Start}$$

(Or Late Finish - Early Finish)

ZERO FLOAT



CRITICAL PATH

Activity is on the critical path. **No room to slip.**

| Any delay = Project Delay.

POSITIVE FLOAT



BREATHING ROOM

Activity has buffer. It can slip by **[Float] days** without affecting the end date.

PROJECT EXAMPLES



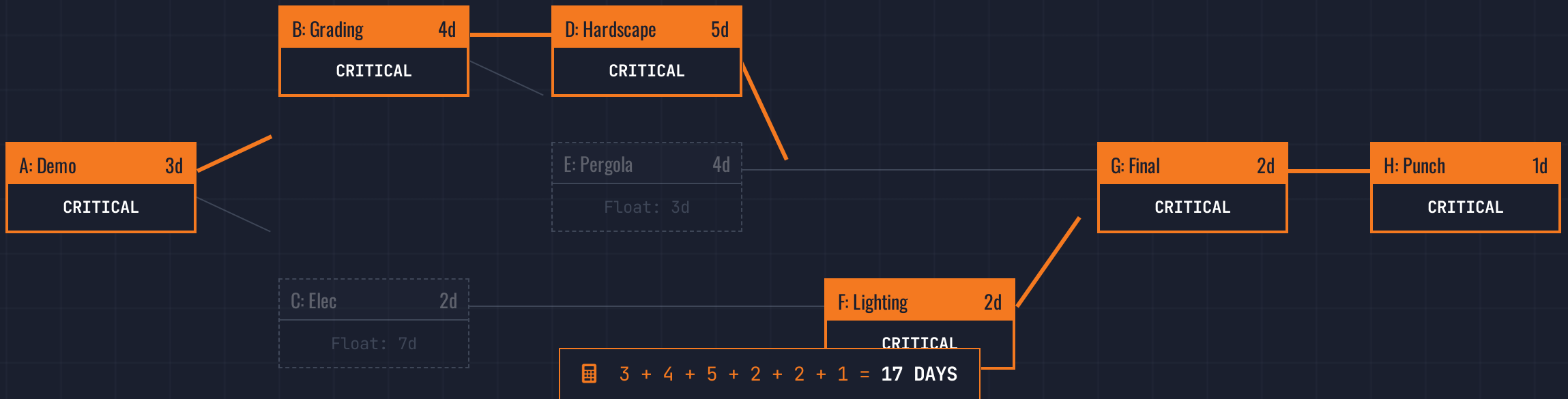
Pergola Build	$10 - 7 = 3 \text{ Days}$
Elec. Rough-In	$10 - 3 = 7 \text{ Days}$
Hardscape	$7 - 7 = 0 \text{ Days}$

HARD TRUTH

PROTECT FLOAT LIKE FINANCIAL MARGIN — ONCE IT'S GONE, EVERYTHING BECOMES CRITICAL.

THE CRITICAL PATH

THE **BACKBONE** OF YOUR PROJECT

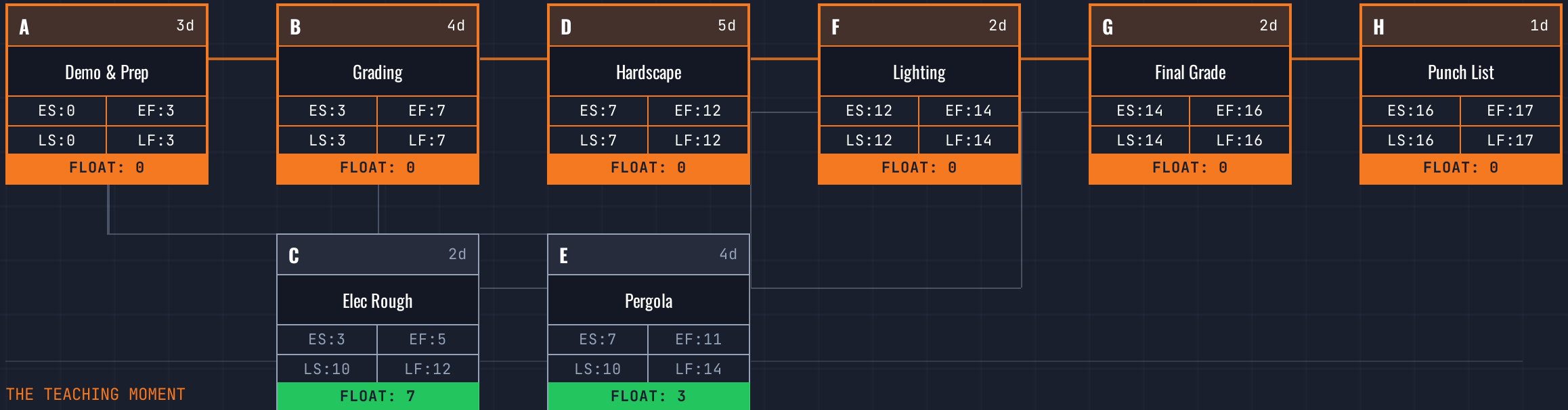


HARD TRUTH

"IF YOU DON'T KNOW YOUR CRITICAL PATH, YOU'RE MANAGING BY GUT. **AND GUT DOESN'T PROTECT MARGIN.**"

PUTTING IT ALL TOGETHER

OUTDOOR LIVING PROJECT — COMPLETE CPM NETWORK



THE TEACHING MOMENT

"Every number here came from **logic and math** — not from a calendar, not from a guess. This is how professionals schedule."

THE SCHEDULE IS A FINANCIAL TOOL

CALLBACK: IT'S NOT JUST A CALENDAR



INPUT

COST LOADING

Every activity has a cost loaded to it. The schedule becomes a cost-over-time model.



OUTPUT

CASH FLOW PROJECTION

Schedule drives when money goes out (subs, materials) and when it comes in. Predictable cash flow = Survival.



REVENUE

BILLING MILESTONES

Schedule completion triggers payment applications. Faster critical path = Faster billing.



RISK

DELAY = MONEY

Schedule delay = Cost escalation + Overhead burn + Lost opportunity on the next job.

HARD TRUTH

"A ONE-WEEK SLIP ON A \$500K JOB DOESN'T COST YOU ONE WEEK. IT COSTS YOU OVERHEAD, REPUTATION, AND THE NEXT JOB YOU CAN'T START."

THE 3-WEEK LOOKAHEAD

YOUR WEEKLY WEAPON

WHAT IT IS

A rolling 3-week window pulled directly from the CPM schedule. Updated every week. The bridge between the master schedule and daily execution.

 UPDATED WEEKLY

// LOOKAHEAD_TEMPLATE

STATUS: ACTIVE

- ≡ Activities starting/finishing in next 3 weeks
Responsible Party (Sub/Crew)
- ✂ Required Resources & Materials
- ⚠ Potential Blockers & Risks
- 🔗 Predecessor Status

HOW TO USE IT

WEEKLY COORDINATION

Run your sub meetings from the lookahead, not from memory.

SUB ACCOUNTABILITY

"You agreed to this sequence and these dates. Are you ready?"

CLIENT COMMUNICATION

"Here's what's happening in the next 3 weeks and what we need from you."



"The lookahead is where the CPM schedule becomes a **daily management tool**. Without it, your schedule lives in a file. With it, your schedule lives on the jobsite."

SCHEDULE AS LEVERAGE

IT'S NOT JUST A PLAN, IT'S A WEAPON



WITH SUBCONTRACTORS

DEPERSONALIZE PRESSURE

Stop being the "bad guy." Let the schedule be the bad guy.

"The schedule we both signed starts Tuesday. Miss it and the job slips."



WITH CLIENTS

QUANTIFY DECISIONS

Show the time cost of indecision or changes immediately.

"That wall move pushes the move-in date by 12 days. Proceed?"



IN DISPUTES

LEGAL DEFENSE

A maintained CPM schedule is your best evidence.

"If you didn't schedule and track it, proving the delay is hard."



INTERNAL TEAM

ALIGNMENT

One source of truth. No "I thought we were doing X."

"If it's not on the schedule, we don't have the manpower."

REMOVE EMOTION. **INSERT DATA.**

COMMON MISTAKES

6 WAYS TO DESTROY YOUR MARGIN



RETROACTIVE SCHEDULING

Building the schedule after the job starts. You're already behind if you start late.



SET IT & FORGET IT

Not updating weekly. A schedule that isn't updated is a historical document, not a management tool.



THE SECRET DOCUMENT

Keeping the schedule in the office. If subs don't see it in coordination meetings, it doesn't exist.



THE "BAR CHART" TRAP

Drawing bars on a page without logic ties. If one moves, nothing else moves. That's a drawing, not a schedule.



FLOAT BLINDNESS

Ignoring float until it's gone. Treating non-critical activities like they have infinite time.



CLIENT THEATER

Treating the schedule as a deliverable to placate the owner, rather than a tool to run the job.

HOW TO START

YOU DON'T NEED SOFTWARE. YOU NEED DISCIPLINE.

01



ANALOG FIRST

Start with a whiteboard or sticky notes. Paper forces focus on logic.

02



DEFINE THE DATA

List activities and assign durations. Dates come later.

03



MAP THE LOGIC

Draw the network. Ask "What must happen before this?" for each item.

04



DO THE MATH

Run forward/backward passes manually. Find the Critical Path first.

05



DIGITIZE

Enter it into MS Project/Primavera/Excel. Software is a calculator/printer.

"THE TOOL DOESN'T MAKE THE SCHEDULE. **THE LOGIC DOES.** IF THE LOGIC IS WRONG, THE SOFTWARE JUST HELPS YOU MAKE MISTAKES FASTER."

THE KPI CONNECTION

WHAT GETS MEASURED GETS MANAGED

● BEHIND SCHEDULE

-3
DAYS

SCHEDULE VARIANCE (SV)

BASELINE
MAR 10

ACTUAL
MAR 13

THE ONE NUMBER THAT MATTERS

Forget "percent complete." It's subjective. Schedule Variance is binary. You are either ahead, on time, or behind.

THE "BUSY" TRAP

"We had 20 guys on site today!" **Doesn't matter.** If they weren't working on the Critical Path, you're still 3 days behind.



"IF PMS ARE NOT MEASURED, THEY WILL DEFAULT TO 'BUSY.'
BUSY IS NOT A BUSINESS METRIC."

// THE FINAL CHOICE

YOUR SCHEDULE IS EITHER **A WEAPON** OR A LIABILITY

"Are you building schedules based on logic and math – or
are you just guessing?"

01 // LOGIC

BUILD THE
LOGIC

02 // DISCIPLINE

UPDATE
WEEKLY

03 // FOCUS

PROTECT THE
CRITICAL PATH

04 // RESULT

OWN THE
OUTCOME

// SCHEDULE_CONTROL // END OF SESSION